Ajloun National University · 314341 AI in Cybersecurity and Cloud Computing

Chapter 1 · Lecture 2 of 2

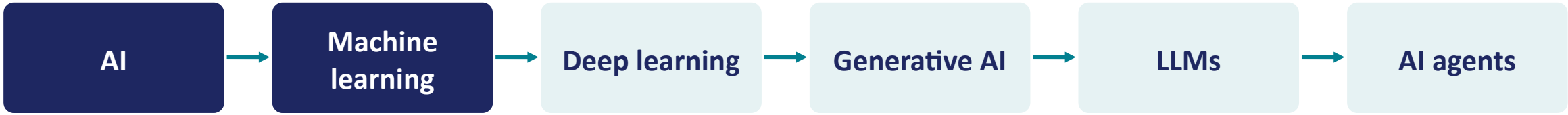
# Introduction to Artificial Intelligence



Deep learning, generative AI, LLMs and AI agents; types, history and uses of AI

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# Recap of Lecture 1




**AI** is a field of computer science that builds machines able to do intelligent tasks.

**Machine learning** learns the rules from examples: data + answers → model.

**Models** such as decision trees, k-NN and k-means do this in different ways.

**Today:** deep learning, generative AI, LLMs and agents; then types, history, domains and cybersecurity.

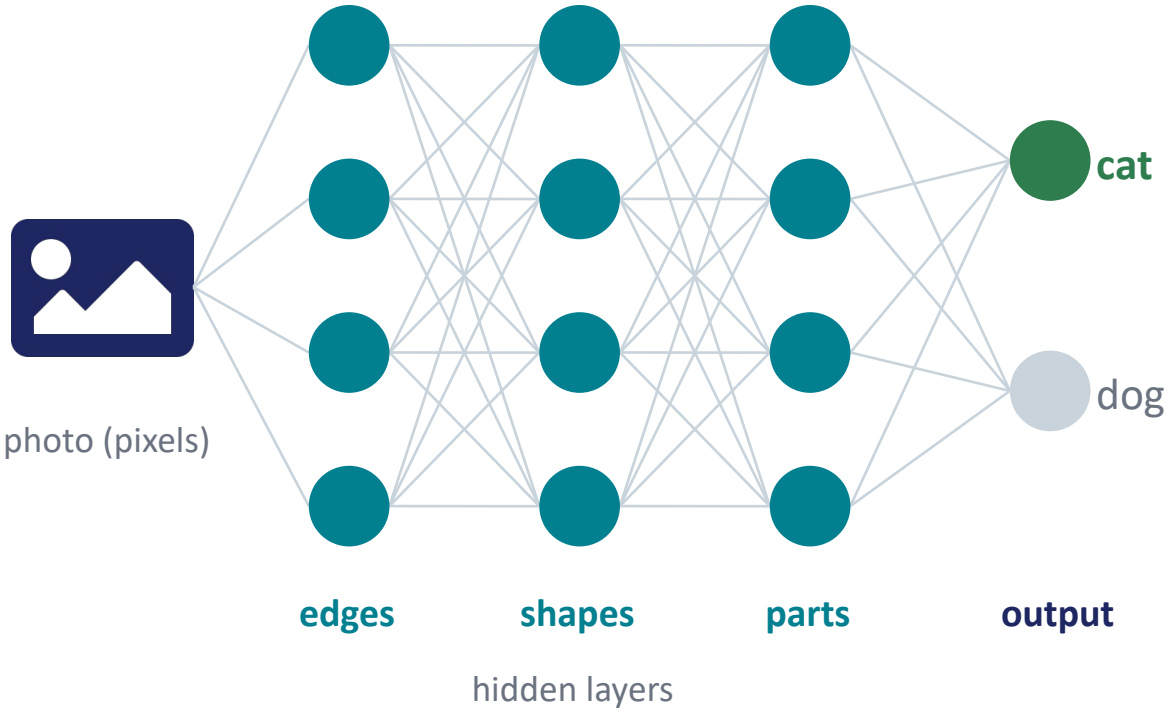
# Deep Learning: Machine Learning with Many Layers

Deep learning is machine learning that uses artificial neural networks with many layers.

**“Deep” means many layers,** not deep understanding.

Each layer turns the data into a more useful form: edges → shapes → parts → object.

**Used in:** face unlock, speech recognition, translation.

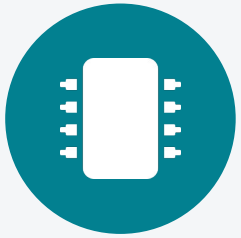


# Why Deep Learning Took Off After 2012



## More data

The internet, phones and sensors produce huge numbers of examples to learn from.



## Faster hardware

Graphics cards (GPUs) made training large networks practical.



## Better methods

Improved training techniques. In 2012 the AlexNet network won the ImageNet image contest.

# Generative AI: From Recognising to Creating

## Predictive AI

chooses between answers

"Is this e-mail spam?"

↓

**spam / not spam**

## Generative AI

creates new content

"Write a polite reminder e-mail to my team."

↓

**a new e-mail, written for you**

**Generative AI** is deep learning that makes new text, images, sound or code after learning from very large collections of examples.

**In security:** defenders use it to write reports; attackers use it to write phishing e-mails.

# Large Language Models (LLMs)

**An LLM** is generative AI for language, trained on huge amounts of text.

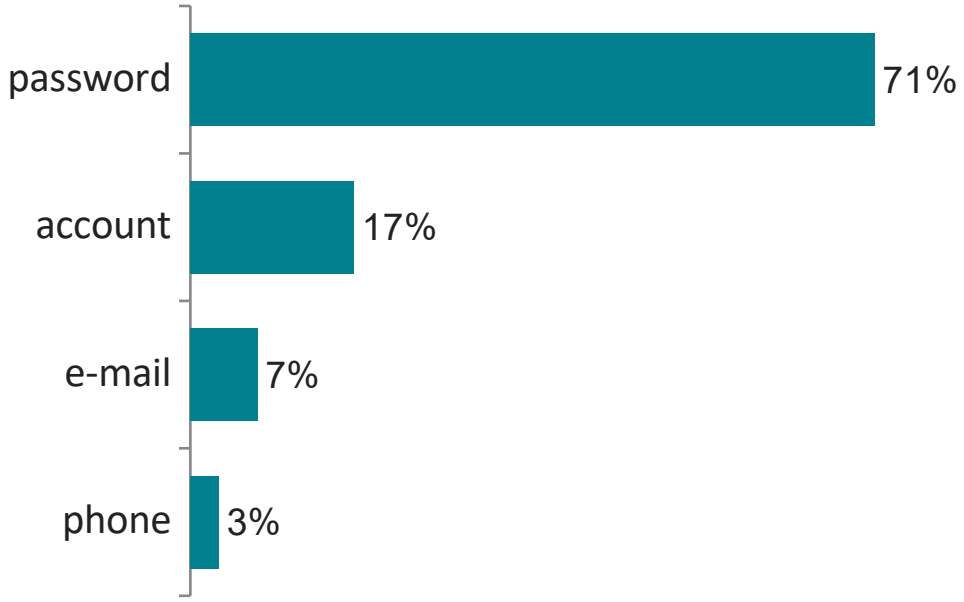
**Its basic job:** predict the next word (token). Repeating this, word after word, produces a full answer.

Most LLMs use the Transformer design (2017). Examples: ChatGPT, Gemini, Claude, Llama.

**Warning:** an LLM can sound sure and still be wrong (“hallucination”). Always check important answers.

**“Click the link to reset your ____”**

**LLM guesses the next word:**




Illustrative values

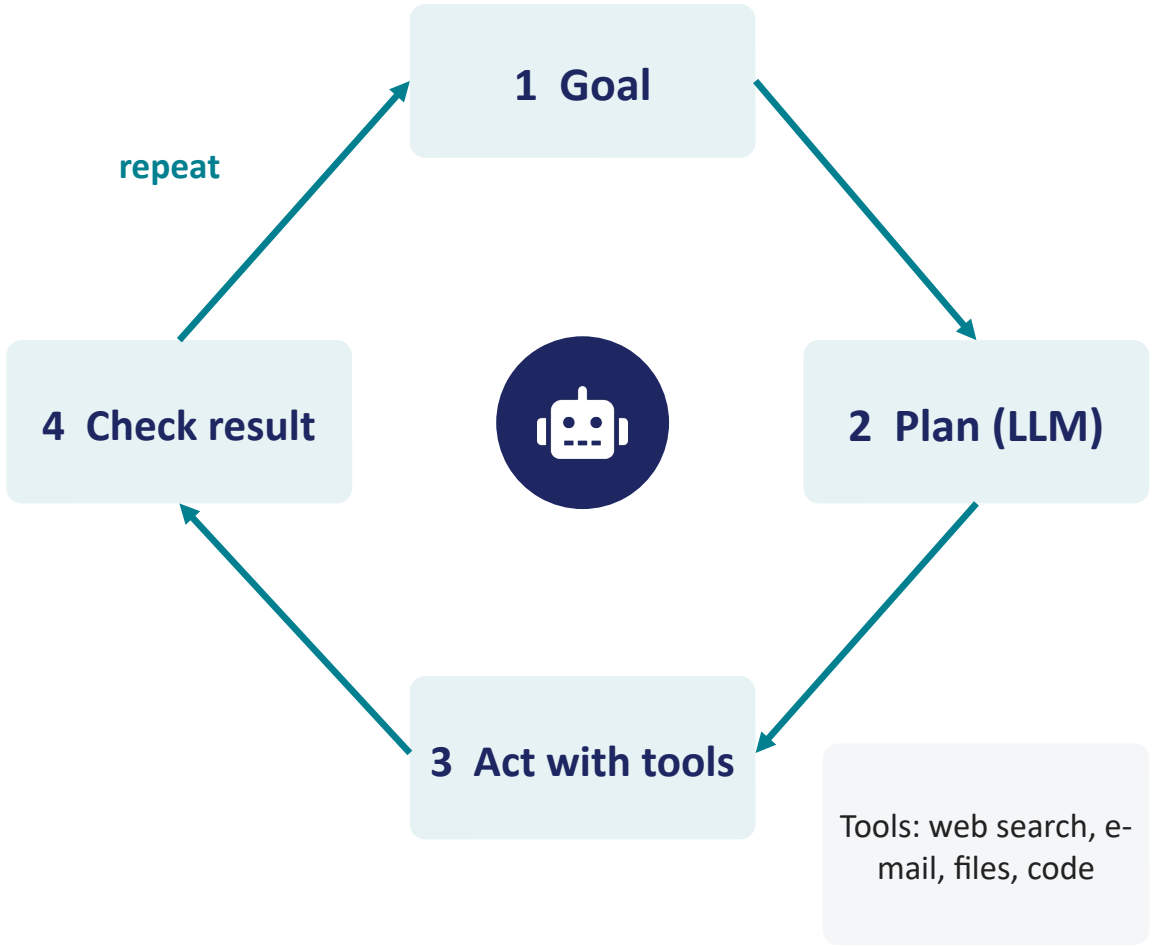
# AI Agents (Agentic AI)

**An AI agent** uses an LLM as its “brain” and adds tools so it can act, not only answer.

Given a goal, it plans steps, uses tools, checks the result and repeats until the goal is reached.

**Security example:** read a new alert → search the logs → look up the IP address → draft a report.

**Rule:** a human approves risky actions.



# The AI Family at a Glance

| Level | What it does | Example |
|---|---|---|
| **Artificial intelligence** | Machines doing tasks that need intelligence | Chess program, spam filter |
| **Machine learning** | Learns the rules from examples | Spam filter trained on old e-mails |
| **Deep learning** | Machine learning with many-layer neural networks | Face unlock, speech recognition |
| **Generative AI** | Creates new text, images, sound or code | Image generators |
| **Large language model** | Generative AI for language (next-word prediction) | ChatGPT, Gemini, Claude |
| **AI agent** | An LLM with tools that acts toward a goal | Assistant that sorts security alerts |

# Types of AI by Capability

more capable

## Narrow AI (ANI)

Does one task, or a limited set of tasks, well.

**Example:** Spam filter, face unlock, chatbots

Exists today

## General AI (AGI)

Would learn and do any intellectual task a person can.

**Example:** No real example yet

Not yet

## Super AI (ASI)

Would be better than humans at almost everything.

**Example:** Science fiction only

An idea

# Types of AI by Functionality

| | | | | |
|---|---|---|---|---|
| 1 | **Reactive machines** | React to the present only; no memory of the past. | Deep Blue chess computer (1997) | Exists |
| 2 | **Limited memory** | Learn from past data to decide better. Most AI today. | Self-driving cars, chatbots | Exists |
| 3 | **Theory of mind** | Would understand people's feelings and intentions. | Research goal | Not yet |
| 4 | **Self-aware AI** | Would have its own awareness and feelings. | None | An idea |

# A Short History of AI

**1950**
Turing asks “Can machines think?”

**1956**
Dartmouth workshop names the field

**1966**
ELIZA, an early chatbot

**1970s**
First “AI winter”: money cut

**1980s**
Expert systems, then a second winter

**1997**
Deep Blue beats chess champion Kasparov

**2012**
AlexNet starts the deep-learning boom

**2016**
AlphaGo beats Go champion Lee Sedol

**2017**
The Transformer design

**2022**
ChatGPT reaches the public

**Pattern: big promises → disappointment (“AI winter”) → a new method → progress**

# Main Branches of AI



## Machine learning

Learning from data (the base of most branches).



## Natural language processing

Understanding and writing human language.



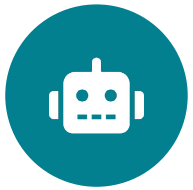
## Computer vision

Understanding images and video.



## Speech

Recognising and producing spoken words.



## Robotics

Machines that sense and act in the physical world.



## Knowledge and planning

Storing facts, reasoning and planning steps.

# AI Across Domains



## Health

Reading X-rays and scans; spotting disease early.



## Finance

Detecting card fraud in real time.



## Transport

Traffic prediction and driver assistance.



## Education

Personal tutoring and automatic feedback.



## Retail

Product recommendations.



## Cybersecurity

Phishing, malware and attack detection.

# AI for Defenders



## Phishing and spam filtering

Learns the clues of fake e-mails.



## Malware detection

Spots harmful files, even new versions.



## Anomaly detection

Flags unusual logins and network traffic.



## Alert triage

Sorts thousands of alerts; agents draft reports.

**USD 1.9 million**

saved per breach, on average, by organisations that use security AI and automation widely

**80 days**

shorter time to find and contain a breach

# AI for Attackers

### Better phishing

AI writes convincing e-mails with no spelling mistakes, in any language.

### Deepfakes

Fake voice or video of a manager asking for a money transfer.

### Faster malware changes

Attackers change code quickly to avoid detection.

**16%** of the data breaches studied by IBM in 2025 involved attackers using AI, most often for phishing and deepfakes.

# Benefits and Risks of AI



## Benefits

- Works fast, 24 hours a day
- Handles millions of events
- Finds patterns people miss
- Helps small security teams



## Risks

- Mistakes: false alarms and missed attacks
- Bias from unfair training data
- Privacy of personal data
- Over-trust in confident wrong answers

**Responsible use: a human stays in charge of important decisions.**

# Activity: AI or Not AI?

**In pairs (5 minutes): mark each item AI or not AI. Reason: does it learn from data or adapt?**

| | | | | | |
|---|---|---|---|---|---|
| 1 | Pocket calculator | AI / not | 5 | Netflix recommendations | AI / not |
| 2 | Gmail spam filter | AI / not | 6 | Excel SUM formula | AI / not |
| 3 | Phone face unlock | AI / not | 7 | Google Maps traffic prediction | AI / not |
| 4 | Traffic light on a fixed timer | AI / not | 8 | ChatGPT | AI / not |

Wrap-up

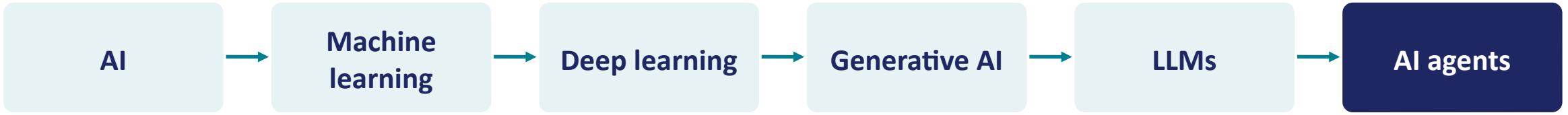
# Key Terms

| | |
|---|---|
| **Artificial intelligence** | The field of making machines do tasks that need intelligence |
| **Agent** | Something that senses its environment and acts on it |
| **Machine learning** | Learning rules from examples instead of writing them |
| **Model** | What machine learning produces: the learned rules |
| **Classification / regression** | Predicting a class (spam?) / predicting a number (price) |
| **Clustering** | Grouping similar items without labels, e.g. k-means |
| **Deep learning** | Machine learning with many-layer neural networks |
| **Generative AI** | AI that creates new text, images, sound or code |
| **Large language model** | Generative AI for language, predicting the next word |
| **AI agent** | An LLM with tools that acts toward a goal |
| **Turing test** | A judge tries to tell a machine from a person by chat |

# Review Questions

1. What is artificial intelligence? Why do we call it a field of computer science?
2. Name the four ways to define AI. Which one do most AI systems follow today?
3. How is machine learning different from classical programming? Name T, E and P for a spam filter.
4. Which model would you use to predict a number? To group computers with no labels? Why?
5. Explain with an example how a decision tree or k-NN decides whether something is an attack.
6. Put in order from widest to narrowest: LLMs, AI, deep learning, machine learning, generative AI.
7. What does an AI agent add to an LLM? Give one security example.
8. Give one way AI helps defenders and one way it helps attackers.

# Summary




**AI** is a field of computer science that builds machines able to do tasks that need intelligence.

**Four definitions;** most AI today follows the rational-agent view: sense, decide, act.

**Machine learning** learns rules from data (task, experience, performance). Models: linear and logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM, k-means.

**Deep learning** uses many layers; generative AI creates; LLMs do it for language; agents add tools to act.

**All AI today is narrow.** General and super AI do not exist.

**In cybersecurity** AI helps both defenders and attackers, so humans must stay in charge.

# Challenge Question (Optional)

**For students who want to go further**

An AI agent can read alerts, search the logs and block IP addresses by itself.

Name two things that could go wrong.

Which rule would you add before letting it run?